

Quantitative Trading — Detailed Summary (Ernest Chan)

Source: Ernest P. Chan, *Quantitative Trading: How to Build Your Own Algorithmic Trading Business* (Wiley; 2nd Edition, Copyright © 2021 by Ernest P. Chan, ISBN 9781119800064) — read from the full text.

Overview & thesis

Chan's book is a practitioner's field manual for turning a systematic trading *idea* into a running *business* — one that a single person can operate from "a spare bedroom." He is explicit that this is not a book about exotic derivatives or advanced math: "The kind of quantitative trading I focus on is called statistical arbitrage trading. Statistical arbitrage deals with the simplest financial instruments: stocks, futures, and sometimes currencies. One does not need an advanced degree to become a statistical arbitrage trader" (Ch. 1). His own credential story is a warning against complexity: PhD from Cornell physics, IBM research, "Morgan Stanley, Credit Suisse, and so on... Hundreds of millions of dollars of trades later, what was the result? Losses, more losses, and losses as far as the eye can see... Finally, I quit... set up a spare bedroom in my home as my trading office, and started to trade the simplest but still quantitative strategies I know... For the first time in my life, my trading strategies became profitable" (Ch. 1). The recurring maxim, attributed to Einstein: "Make everything as simple as possible. But not simpler."

The book's throughline is a discipline of skeptical honesty. A trading idea is worth nothing until it survives a *bias-free, cost-inclusive* backtest, is judged by *risk-adjusted* return, is proven on *out-of-sample* data long enough to be statistically real, and is sized by an *explicit risk rule* (Kelly). The governing rule of thumb: "any strategy that has a Sharpe ratio of less than 1 is not suitable as a stand-alone strategy" (Ch. 2). And on the closing question — can a retail trader still make money? — Chan answers yes, but for a specific structural reason: **capacity**. "It is far, far easier to generate a high Sharpe ratio trading a \$100,000 account than a \$100 million account... This is the niche for independent traders like us" (Ch. 8).

How the book is organized

Eight chapters, each a stage of building the business: - **Ch. 1 — The Whats, Whos, and Whys of Quantitative Trading:** what quant/algorithmic trading is (and isn't), who can do it, the business case (scalability via leverage, low overhead), and the temperament required — "emotion has found the right balance between fear and greed." - **Ch. 2 — Fishing for Ideas:** where strategies come from (blogs, academic papers, newspapers), and the *screening questions* to ask of any candidate — Sharpe, drawdown depth/length, transaction costs, survivorship bias, performance decay, data-snooping, and capacity. - **Ch. 3 — Backtesting:** the technical core — platforms (Excel, MATLAB, Python, R), data sources and quality, the three headline performance measures, statistical significance, the three deadly biases (look-ahead, survivorship, data-snooping), out-of-sample testing, and transaction costs, all with worked code examples. - **Ch. 4 — Setting Up Your Business:** retail vs. proprietary firm, brokerage selection, incorporation/liability, physical infrastructure. - **Ch. 5 — Execution Systems:** automated trading

systems, minimizing market-impact/slippage, paper trading, and diagnosing live-vs-backtest divergence. - **Ch. 6 — Money and Risk Management:** the Kelly formula, optimal leverage and capital allocation, half-Kelly, stop-losses, and trading psychology (despair and greed). - **Ch. 7 — Special Topics:** mean-reversion vs. momentum, stationarity/cointegration/pairs, Conditional Parameter Optimization (the new ML method), factor models, seasonal and high-frequency strategies, and exit design. - **Ch. 8 — Conclusion:** why independent traders can beat institutions (capacity, freedom from constraints, incentives), and the reality of alpha decay.

The 2nd-edition preface flags the new material: "a novel application of machine learning" for regime adaptation (CPO), Python/R/MATLAB code throughout, "new Python/R codes on checking for stationarity and cointegration, and some surprising out-of-sample results on seasonal trading."

Core concepts (deep)

Judge by risk-adjusted return, not raw return. "I would argue that the Sharpe ratio, maximum drawdown, and MAR ratio are the most important" (Ch. 3). Chan deliberately excludes CAGR as a headline number because it is ambiguous about the denominator (one side or both sides of a long-short book? leveraged or unleveraged?), whereas the Sharpe triplet is comparable across strategies and traders.

Drawdown is the emotional and business reality of risk. "A drawdown at a given time t is defined as the difference between the current equity value... and the global maximum of the equity curve occurring on or before time t ." The peak is the "high watermark"; maximum drawdown duration "is the longest it has taken for the equity curve to recover losses" (Ch. 2). Chan is candid: "in real life there is nothing more gut-wrenching and emotionally disturbing to suffer than a drawdown." He urges you to compare a strategy's backtested drawdown against your own tolerance ("Would it be 20 percent and three months, or 10 percent and one month?") *before* trading.

Backtest honesty is the whole game. Three biases silently inflate results: - *Survivorship bias* — databases that drop bankrupt/delisted/merged names. "Backtesting a strategy using data with survivorship bias can be dangerous because it may inflate the historical performance... This is especially true if the strategy has a 'value' bent" (Ch. 2). It hurts *older* periods most, "because the further back we go... the more missing stocks we will have." - *Look-ahead bias* — "using information that was available only at a time ahead of the instant the trade was made" (Ch. 3). The cure is mechanical: lag every signal. - *Data-snooping bias* — overfitting parameters to "historical accidents... that will not repeat." "Simple models are often the ones that will stand the test of time" (Ch. 2).

Costs are decisive, not a footnote. "No backtest performance is realistic without incorporating transaction costs" (Ch. 3). Costs are commission + liquidity cost (bid-ask) + opportunity cost (unfilled limits) + market impact + slippage.

Sizing is separate from, and as important as, the signal. "Risk always decreases long-term growth rate — hence the importance of risk management!" (Ch. 6). The Kelly formula tells you the leverage that maximizes long-term compounded wealth without risking ruin.

Markets are non-stationary; regimes shift. "The market is not stationary; why should your strategies be?" (Preface). This is why more data is *not* always better — "This is true only when the financial time

series is generated by a stationary process. Unfortunately, financial time series is famously nonstationary" (Ch. 2) — and why he demands strong performance on *recent* data.

Key frameworks, metrics & formulas

Sharpe ratio. Annualized mean excess return $\div$ annualized standard deviation of returns; a special case of the information ratio. Annualization: multiply the per-period Sharpe by $\sqrt{N_T}$, where N_T is the number of periods per year (252 daily; $252 \times 6.5 = 1,638$ hourly — "A common mistake is to compute N_T as $252 \times 24 = 6,048$ "). Subtlety: for a dollar-neutral (self-financing) or intraday long-only book you do **not** subtract the risk-free rate. Rules of thumb (Ch. 2): "For a strategy that achieves profitability almost every month, its (annualized) Sharpe ratio is typically greater than 2. For a strategy that is profitable almost every day, its Sharpe ratio is usually greater than 3." Worked (Example 3.4): buy-and-hold IGE, 2001–2007, gives Sharpe **0.789317538**; a long-IGE/short-market-neutral hedge gives **0.783681**.

Statistical significance of the Sharpe ratio (Ch. 3, from Bailey et al.). Three benchmarks quoted verbatim: - "If you want to be statistically confident (at the 95 percent level) that your true Sharpe ratio is equal to or greater than 0, you need a backtest Sharpe ratio of 1 and a sample size of **681 data points** (e.g., **2.71 years** of daily data)." - "If your backtest Sharpe ratio is 2 or more, then you need only **174 data points** (0.69 years)... to be confident that your true Sharpe ratio is... greater than 0." - "If you want to be confident that your true Sharpe ratio is... greater than 1, then you need a backtest Sharpe ratio of at least **1.5** and a sample size of **2,739** (10.87 years)." Crucially: "these results not only apply to backtest, but also to out-of-sample (paper trading) length."

Maximum drawdown & MAR ratio. Max drawdown = peak-to-subsequent-trough equity decline (time order matters). "MAR ratio is just the ratio of the CAGR and the maximum drawdown." Worked (Example 3.5): max drawdown **10.53 percent**, max drawdown duration **497 trading days**. Note "The maximum drawdown and the maximum drawdown duration do not typically overlap."

Deflated Sharpe Ratio. Bailey (2014) — deflates your backtest Sharpe by "how many times you have tweaked a backtest to obtain that amazing Sharpe ratio." Chan's parameter budget: "I would not employ more than **five parameters**."

Kelly optimal leverage (Ch. 6). Full vector form for n strategies: $\mathbf{F}^* = \mathbf{C}^{-1}\mathbf{M}$ (inverse covariance matrix $\times$ mean-return vector), where returns are "one-period, simple (uncompounded), unlevered." For independent strategies $\mathbf{C}$ is diagonal and it collapses to the famous scalar $f = m/\sigma^2$ (mean $\div$ variance). SPY worked example (Example 6.2): mean annual return 11.23%, annualized σ 16.91%, risk-free 4% $\rightarrow$ excess return 7.231%, Sharpe 0.4275, and "The optimal leverage according to the Kelly formula is $f = 0.07231/0.1691^2 =$ **2.528**." Levered compounded growth = **13.14%**; unlevered growth $g = r + m - \sigma^2/2 = 0.1123 - 0.1691^2/2 =$ **9.8%** — "and not mean annual return of 11.23 percent." The maximum growth rate under Kelly is proportional to the Sharpe ratio squared. Derivation (appendix): maximize $g(f) = r + fm - f^2\sigma^2/2$, set $dg/df = 0 \rightarrow$ " $f = m/\sigma^2$, the Kelly formula."

Why risk itself is a cost (Example 6.1). A stock going $\pm 1\%$ each minute (50/50) does not break even — "you will lose money, at the rate of 0.005 percent (or 0.5 basis point) every minute," because compounded growth is $g = m - \sigma^2/2$, and the geometric mean is below the arithmetic mean.

Half-Kelly and the stress cap. "Because of uncertainties in parameter estimations, and also because return distributions are not really Gaussian, traders prefer to cut this recommended leverage in half for safety. This is called half-Kelly betting." Beyond that, use a fat-tail stress cap: "The leverage to use is always the smaller of the half-Kelly leverage and the maximum leverage obtained using the worst historical loss." Worked: S&P 500 worst one-day loss was **-20.47% on October 19, 1987 ("Black Monday")**; if you tolerate a 20% one-day drawdown, max leverage ≈ 1 , yet half-Kelly recommends **1.26** — "even half-Kelly leverage would not be conservative enough to survive Black Monday."

Transaction costs, quantified. For S&P 500 stocks "the average transaction cost (excluding commissions)... would be about **5 basis points**"; a round trip is two trades = 10 bps. For ES (E-mini futures) "about **1 basis point**." Two blow-up examples: (1) a 5-minute Bollinger-Band mean-reverter on ES has "a Sharpe ratio... about **3** without transaction costs... reduced to **-3** if we subtract 1 basis point" (Ch. 2). (2) The Khandani-Lo mean-reversal strategy (buy worst prior-day losers, short best winners) had a Sharpe of **4.47 in 2006**, becoming "an unprofitable Sharpe ratio of **-3.19** after transaction costs" of 5 bps/trade (Example 3.7).

Half-life of mean reversion (Ornstein-Uhlenbeck, Example 7.5). Fit $dz = \theta(z - \mu)dt + dW$; "the half-life of this exponential decay is equal to $\ln(2)/\theta$," a statistically robust way to set the holding period for a mean-reverting position. GLD-GDX spread half-life $\approx$ **7.84 days** (Python) / ~ 10 days.

Chapter / thematic walkthrough

Ch. 1 — Is this business for you? Quant trading = "trading of securities based strictly on the buy/sell decisions of computer algorithms." It subsumes but exceeds technical analysis and can ingest fundamentals and even parsed news. Its distinctive business feature is scalability: "scaling up often just means changing a number in your program. This number is called leverage." The ideal trader "has enough savings to withstand the inevitable losses and periods without income," because "strategies have intrinsic rates of returns that cannot be hurried."

Ch. 2 — Fishing for ideas frames the seven questions to interrogate any candidate strategy: How high is the Sharpe (rules of thumb above)? How deep/long the drawdown ("more than 10 percent" deep or "four or more months" long is a red flag)? How much do transaction costs bite? Is the data survivorship-free? Did performance decay over the years (favor recent)? Is there data-snooping (fewer parameters)? What is its capacity? A memorable line: infrequent traders "won't have a high Sharpe ratio... but it does disqualify the strategy from being your main profit center."

Ch. 3 — Backtesting is the technical heart. On platforms: Excel is WYSIWYG and resistant to look-ahead bias but limited; MATLAB and especially **Python** (numpy/pandas, scikit-learn) dominate for portfolio-scale work; the book ships code in all of MATLAB/Python/R. Data quality warnings: "the high and low prices are far noisier than the open and close" (a limit-order fill at the recorded high may never have happened), so backtests leaning on highs/lows "almost always inflate your backtest returns." The survivorship demonstration (Example 3.3, a "buy low-priced stocks" toy) is the book's most vivid number: the true return picking the 10 cheapest of the top-1,000 stocks at the start of 2001 was **-42 percent**; the same rule on a survivorship-biased database that quietly dropped the delisted names returned **+388 percent** — "a fictitious return that was due to survivorship bias in our database." Look-ahead examples: "Buy when the

stock is within 1 percent of the day's low" (you couldn't know the low until the close), or fitting a regression on the *whole* dataset to trade its middle. The cure: "Use lagged historical data... based on data up to the close of the previous trading period only." And the **A-vs-B truncation test**: run the backtest on full data → positions file A; truncate the last N days (N = 10 to 100) → positions file B; the overlapping rows must be identical, else "you are inadvertently using the truncated part of the historical data... in determining the positions in file A." Out-of-sample: split older→recent (train→test), size the test window by the same 681-point rule, and prefer "moving optimization" / "parameterless" models. The chapter ends on transaction costs (Example 3.7 above).

Ch. 4 — Setting up your business. Retail account (Reg-T leverage ~2× overnight; unlimited personal liability) vs. proprietary firm (Series 7, higher leverage, profit split, liability limited to your deposit). Box 4.1 strongly recommends incorporating (S-corp/LLC) to cap liability — motivated by real cases like the Jan-2015 Swiss-Franc unpegging where brokers pursued clients for negative equity.

Ch. 5 — Execution systems. Build an automated trading system (ATS) to pull data, generate and route orders. Fully automated is "indispensable" for high-frequency; low-frequency traders can go semi-automated (Excel/MATLAB generating a basket). To minimize market impact, "limit the size (number of shares)... scale the size of your orders," and use limit orders (which trade impact for slippage/opportunity cost). Paper trading is the "ultimate out-of-sample test" and often the only way to "discover look-ahead errors." When live diverges from backtest, the "two most dreaded causes" are "data-snooping bias and regime shift" — e.g., decimalization and the June-2007 removal of the short-sale plus-tick rule.

Ch. 6 — Money and risk management develops Kelly (formulas above), then adds the human layer: the two enemies are **despair** (shutting a good model in a drawdown, or doubling down) and **greed** (over-leveraging a hot model). Both cause over-leverage, "the one golden rule in risk management is to keep the size of your portfolio under control." Chan cites LTCM (2000) and Amaranth Advisors (2006, a "\$6 billion loss... textbook case of risk mismanagement") and confesses his own two over-leverage disasters (a >\$1M institutional loss adding \$100M to a six-month-old strategy; a near-six-figure personal loss on an XLE/CL spread that "started to revert afterward when I wasn't around to benefit"). On stop-losses, the nuanced verdict: "It is a common fallacy to believe that imposing stop loss will prevent the portfolio from suffering catastrophic losses... For stop loss to be beneficial, we must believe that we are in a momentum, or trending, regime" — news-driven moves are momentum (don't "stand in front of a freight train"); newsless liquidity events tend to mean-revert (stops harmful).

Ch. 7 — Special topics. The two families: "Trading strategies can be profitable only if securities prices are either mean-reverting or trending. Otherwise, they are random-walking, and trading will be futile." Time-series mean reversion (single stock) "doesn't happen often"; cross-sectional mean reversion (spreads/portfolios) "happens much more often" — but its backtests are uniquely vulnerable to bad ticks and survivorship. Momentum arises from slow information diffusion (→ PEAD, post-earnings-announcement drift), incremental large-order execution, and herding. **Stationarity & cointegration**: a stationary series is "integrated of order zero, or I(0)" and an ideal mean-reversion candidate; most single stocks aren't, but a pair can be cointegrated — e.g., long 1 GLD, short ~1.63 GDX yields a stationary spread, tested by the "cointegrating augmented Dickey-Fuller test" (CADF).

Conditional Parameter Optimization (CPO): the flagship new ML method — instead of one fixed parameter set (Unconditional/Walk-Forward optimization), train "random forest with boosting" on ~115

features (technical indicators over many lookbacks) plus candidate parameter values to predict each parameter set's next-day return, then pick the best set *each day*; demonstrated on a Bollinger-Band GLD strategy. **Factor models (APT):** $R = Xb + u$; illustrated with Fama-French (market, SMB = small-minus-big, HML = high-minus-low value), plus WML momentum. Key caution: factor regressions are "contemporaneous, not predictive" — usable for trading only because "factor returns are more stable... they have momentum." A good factor model's R^2 (1,000 stocks, 50 factors, monthly) "is typically about 30 to 40 percent." **Exits:** four types — fixed holding period, target/profit cap, latest entry signal, stop price; for mean-reverters use the Ornstein-Uhlenbeck half-life. **Seasonal:** the January effect on S&P 600 small-caps "did not work in 2006–2007, but worked wonderfully in January 2008" — much equity seasonality has decayed, though some commodity-futures seasonals persist.

Ch. 8 — Can independent traders succeed? Yes, because of **capacity**: small accounts can act as market makers "providing short-term liquidity... and taking quick profits," a niche too small for billion-dollar funds, which must instead demand liquidity, hold long, and absorb regime shocks. Independents are also free of institutional constraints ("any constraint imposed on an optimization problem decreases the optimal objective value") and misaligned incentives. But alpha decays — "strategies do lose their potency over time... It takes ongoing research."

How it maps to the Trader Screener

1. **§2g "Max drawdown (historical)"** is one of Chan's three headline risk measures ("the Sharpe ratio, maximum drawdown, and MAR ratio are the most important," Ch. 3). The screener should compute it — plus MAR ($\text{CAGR} \div \text{maxDD}$) and Sharpe — on the equity curve of any ranked/composite strategy, and flag drawdowns deeper than ~10% or longer than ~4 months as Chan's Ch. 2 red lines, so ranking is on survivability, not raw return.
2. **§2g "Suggested position size (ATR-based)"** is the operational form of Kelly $f = m/\sigma^2$: risk scales *inversely* with variance. Because ATR (**§2b "ATR / ATR%"**) is the screener's practical volatility proxy, divide a **half-Kelly** risk budget by ATR to get share count — never full Kelly, per "traders prefer to cut this recommended leverage in half for safety," and cap it further by the worst-historical-loss stress rule ("the smaller of the half-Kelly leverage and the maximum leverage obtained using the worst historical loss," Ch. 6).
3. **§2g "Stop distance (ATR multiple)"** implements Chan's exit discipline (Ch. 7) but with his caveat: "For stop loss to be beneficial, we must believe that we are in a momentum, or trending, regime." Gate ATR stops on **§2a "Moving averages (50/200d) + cross"** (trend confirmation) and don't apply them to mean-reverting names, where a stop "realizes the catastrophic loss and not avoids it."
4. **Look-ahead & survivorship guardrails on every §2a/§2b indicator + universe.json.** "Return 1/3/6/12-month," "Relative strength rank (vs universe)," MA cross, "RSI," "MACD," **§2b "ATR / ATR%," "Realized (historical) volatility," "Beta (vs SPY)"** must all be computed "based on data up to the close of the previous trading period only" — lag every signal. And **universe.json must retain delisted/merged/bankrupt names (point-in-time)**, or momentum/value backtests inflate exactly as in Example 3.3 (–42% real vs +388% survivorship-inflated). Validate with Chan's A-vs-B truncation test (N = 10–100 days) to catch subtle leakage.

5. **§2d "Composite score (model.json)"** is Chan's ranked-factor composite (Fama-French-style, Ch. 7): keep parameters ≤ 5 , validate out-of-sample (older = train, recent = test, test-window length set by the 681-point / backtest-Sharpe ≥ 1 rule, not a fixed 1/3), and remember factor exposures are contemporaneous — usable only if the factor return has momentum. Treat **§2d "Upside % (DCF/comps)"** and any fundamental factor as point-in-time: fundamental databases carry "embedded look-ahead bias" from restated financials, and a "value bent" maximizes survivorship inflation.
6. **§2f macro (VIX / rates)** is Chan's regime filter. Since "financial time series is famously nonstationary," condition rankings on *recent* performance and use VIX/yield-curve state to switch strategy family (mean-reversion vs. momentum) or at least parameters — a lightweight stand-in for CPO's per-day parameter adaptation.
7. **§2c liquidity (ADV, dollar volume)** operationalizes Chan's capacity and market-impact themes (Ch. 5, Ch. 8): screen out names too thin to trade at your size without slippage, and treat capacity as a first-class filter — the independent trader's edge lives specifically at low capacity.
8. **Transaction-cost realism across all sorts.** Before trusting any composite ranking, net out ~ 5 bps/trade for stocks (10 bps round trip); Chan's Sharpe-3 $\rightarrow -3$ and 4.47 $\rightarrow -3.19$ examples show a ranking can invert entirely once costs are applied.

Actionable takeaways

- Attach **Sharpe + max drawdown + MAR** to §2d before trusting any ranking; require backtest Sharpe ≥ 1 over ≥ 681 daily points (≥ 1.5 over $\geq 2,739$ points to trust a true Sharpe ≥ 1).
- **Lag every §2a/§2b indicator** to the prior close; point **universe.json** at a delisting-inclusive (point-in-time) source; run the **A/B truncation check** on any backtest.
- Size **§2g** from **half-Kelly (m/σ^2)** translated through **§2b ATR**, then cap by the worst-historical-loss rule; gate **§2g ATR stops** on **§2a trend/regime**.
- Split history older $\rightarrow$ recent, hold out per the sample-size rule, **paper trade**, keep parameters ≤ 5 , and **always net out realistic transaction costs** (≈ 5 bps/trade equities).
- Use **§2f VIX/rates** as a regime switch (mean-reversion vs. momentum) and weight recent data because markets are non-stationary.
- Add a **capacity / liquidity filter (§2c)** — the independent edge is at low capacity; don't screen for strategies only a large fund could run.

Caveats, limits & what to distrust

- **The Gaussian assumption behind Kelly is wrong in the tails.** Chan says so plainly: large losses "occur at far higher frequencies than would be predicted by a nice bell-shaped curve" (black-swan/fat-tail events). Hence half-Kelly *and* the historical-worst-loss cap — and even those "would not be conservative enough to survive Black Monday."

- **Estimates are noisy.** Kelly's m and σ (and Sharpe, drawdown) are sample estimates that drift; Chan re-estimates leverage continuously on a trailing window and lowers it toward zero as a model decays.
- **Backtest numbers are upper bounds.** Where he skips survivorship-free data (Example 3.7) he warns the results "are upper bounds on the actual performance." Mean-reversion backtests are *especially* inflated by bad ticks and survivorship.
- **Non-stationarity limits data value.** "Only data within the past 10 years are really suitable for building [a] predictive model," and regime shifts can obsolete even that. More history $\neq$ more robustness.
- **Alpha decays.** Strategies "lose their potency over time as more traders catch on"; seasonal equity effects have largely faded.
- **Specifics are dated (2021 edition).** Broker names (Alpaca, Interactive Bishop/IB), APIs, PredictNow.ai, and cost figures are illustrative; treat the *method* as durable and the vendor/number specifics as needing refresh.
- **CPO is proprietary-tinged.** The ML "Conditional Parameter Optimization" is demonstrated through Chan's own PredictNow.ai API; the concept (per-regime parameter selection) is reusable, but the specific results should be read with the author's commercial interest in mind.

Open questions

- Which **delisting-aware / point-in-time** data source feeds a survivorship-clean **universe.json** at startup cost?
- Do we size **\$2g** by *true* Kelly (m/σ^2 across the whole book, via $C \blacksquare^4 M$) or the simpler ATR fixed-fractional column — and at what fraction (half-Kelly, or the worst-loss cap when tighter)?
- Are **point-in-time fundamentals** available for \$2d/DCF inputs, or is some look-ahead unavoidable in restated financials?
- What **rolling lookback and re-estimation cadence** for Sharpe / drawdown / Kelly, and what concrete VIX/rate thresholds define the \$2f regime switch?
- Can a lightweight version of **CPO** (per-regime parameter selection) be justified for a nightly screener, or does its data/infra cost exceed the edge at our capacity?

Sources

- **Primary:** Ernest P. Chan, *Quantitative Trading: How to Build Your Own Algorithmic Trading Business*, 2nd Edition, John Wiley & Sons, © 2021 (ISBN 9781119800064) — summarized from the full text.
- **Chapters referenced:** Ch. 1 (Whats/Whos/Whys — statistical arbitrage, simplicity, business case, leverage/scalability); Ch. 2 (Fishing for Ideas — Sharpe rules of thumb, drawdown definitions, transaction-cost/Bollinger-ES example, survivorship & data-snooping intros, non-stationarity, capacity); Ch. 3 (Backtesting — platforms/data, Sharpe/maxDD/MAR, Examples 3.3–3.7, statistical-significance 681/174/2,739 points, look-ahead A/B truncation test, out-of-sample split, transaction costs); Ch. 4 (Setting Up Your Business — retail vs. prop, incorporation/liability); Ch. 5 (Execution Systems — ATS, market

impact/slippage, paper trading, regime-shift divergence); Ch. 6 (Money & Risk Management — Kelly $F^* = C \sqrt{M}$ and $f = m/\sigma^2$, SPY Example 6.2, half-Kelly, Black-Monday cap, Example 6.1 risk-drag, stop-loss regime caveat, LTCM/Amaranth, Kelly derivation appendix); Ch. 7 (Special Topics — mean-reversion vs. momentum, stationarity/cointegration/CADF, Ornstein-Uhlenbeck half-life, Conditional Parameter Optimization, Fama-French factor models, January effect, exits); Ch. 8 (Conclusion — capacity, institutional constraints, alpha decay).